Electrostatics

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Coulomb's Law & Electric Field

Source charges q — responsible for creating electric field/force

Test charge Q — charge used to measure the exisiting electric field/force

Principal of Superposition — interaction between two charges is independant of other charges

Coulomb's Law

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} \tag{1}$$

Where $\vec{\boldsymbol{\mathscr{r}}} = \vec{r} - \vec{r'}$ is the **separation vector**, $\vec{r}$ & $\vec{r'}$ are **position vectors** for test charge and source charge respectively

Permitivity of free space $\epsilon_0 \equiv 8.85 \times 10^{-12}\frac{C^2}{N \cdot m^2}$

The Electric Field

Force due to multiple charges,

$$\vec{F} = \vec{F}_{total} = \vec{F}_1 + \vec{F}_2 = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 Q}{\mathscr{r}_1^2}\hat{\boldsymbol{\mathscr{r}}}_1 + \frac{q_2 Q}{\mathscr{r}_2^2}\hat{\boldsymbol{\mathscr{r}}}_2 + \dots\right)$$
$$\vec{F} = \frac{Q}{4\pi\epsilon_0}\left[\frac{q_1}{\mathscr{r}_1^2}\hat{\boldsymbol{\mathscr{r}}}_1 + \frac{q_2}{\mathscr{r}_2^2}\hat{\boldsymbol{\mathscr{r}}}_2 + \dots\right]$$

$\vec{E} \equiv$ electric field of source charges.

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{\mathscr{r}_1^2}\hat{\boldsymbol{\mathscr{r}}}_1 + \frac{q_2}{\mathscr{r}_2^2}\hat{\boldsymbol{\mathscr{r}}}_2 + \dots\right] = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\mathscr{r}_i^2}\hat{\boldsymbol{\mathscr{r}}}_i \tag{2}$$

then,

$$\vec{F} = Q\vec{E}(\vec{r}) \tag{3}$$

$\vec{E}(\vec{r}) \equiv$ force per unit charge on a test charge at point P

Continuous Charge Distributions